Tensorial Multiview Subspace Clustering for Polarimetric Hyperspectral Images

Zhengyi Chen, Chunmin Zhang, Tingkui Mu, and Yifan He

***Abstract*— Polarimetric hyperspectral images (PHSIs) can provide complementary representations of a scene from the perspectives of images, spectra, and polarization at the same time and are expected to improve the quality of scene description. In this article, the clustering for PHSI is deemed to be a multiview clustering task, and a tensorial polarimetric-spectral multiview subspace clustering (TPS-MSC) algorithm for PHSI is proposed. It constructs a small size dictionary, instead of a large self-representative dictionary, by pre-clustering each view independently to give a sparse representation of all the data. Then, the view-specific representation matrices are tensorized to explore the low-rank structure among different views, and the consistency of all views in pre-clustering is incorporated into the representation learning framework to strengthen the interview correlations. The proposed model is efficiently optimized by the alternative direction minimization of multipliers (ADMM) algorithm. Some experiments are carried out to validate the capacity of PHSI for target identification and to demonstrate the accuracy and efficiency of the proposed TPS-MSC algorithm.**

***Index Terms*— Low-rank tensor, multiview subspace clustering, polarimetric hyperspectral images (PHSIs), subspace learning.**

I. Introduction

SPECTRA and polarization are two of the fundamental optical properties of objects. Hyperspectral images (HSIs) consisting of rich spatial correlation and high-resolution spectral composition are commonly used in applications requiring fine identification of materials or precise estimation of physical parameters [1], [2]. Polarization that conveys surface features including shapes, shadows, edges, and roughness can provide information complementary to HSI [3], [4]. As a combination of HSI and polarization, polarimetric HSIs (PHSIs) containing four Stokes vectors under all hyperspectral bands are expected to have great application potential in various tasks, such as object detection, supervised classification, and unsupervised clustering.

Compared with the standard HSIs, PHSIs have an additional polarimetric dimension containing multiple polarimetric directions or four Stokes vectors, which means that PHSI can describe more information about the target. However, this additional dimension also poses difficulties in data acquisition, usually requiring sacrificing the spatial and spectral resolution of the instrument. With the development of imaging spectropolarimeters, such as the variable-retardance Fourier-transform imaging spectropolarimeter [5] and the channeled interference imaging spectropolarimeter (CIISP) [6], PHSI has become readily available and has been validated for remote sensing applications. Cheng and Reys [7] evaluated PHSI's ability to describe different artificial and natural objects and applied PHSI to the differentiation of minerals. Through a series of outdoor experiments, Bartlett and Gartley [8] have successfully used the difference between targets and backgrounds exhibited on PHSI to detect and track moving targets. van Harten and de Boer [9] developed a ground-based spectropolarimeter that has been applied for atmospheric aerosol characterization. By an airborne imaging spectropolarimeter, Diner and Xu [10] revealed the potential of PHSI for aerosol and cloud remote sensing.

There have been many studies on HSI clustering [11]–[18], but most of them suffer from significant misclassification on HSI clustering tasks because of the large spectral variability, high dimensionality, and complex spatial structures. Recently, assuming that pixels of different spectra of a species are located in the same subspace, some subspace clustering algorithms, the most representative of which is the sparse subspace clustering (SSC) algorithm [19], have achieved remarkable success in the HSI clustering task. Based on the unique spectral characteristics of HSI, numerous improved algorithms based on SSC are constantly being proposed, such as the l_2-norm regularized SSC (L2-SSC) [20] and the multiobjective SSC (MOSSC) [21]. Besides, considering the spatial continuity of HSI, several spectral–spatial methods further enhance the clustering performance, such as the spectral–spatial SSC (S^4C) algorithm [22] and the nonlocal means regularized sketched reweighted sparse and low-rank (NL-SSLR) SC algorithm [23]. Furthermore, Cai *et al.* [24] revisited the subspace clustering with graph convolution and presented the graph convolutional subspace clustering (GCSC) algorithm. Huang *et al.* [25] proposed the novel structural subspace clustering (STSC) model, which leveraged the structural prior information and the self-representation property.

The methods mentioned earlier are for single-view HSI data, and the clustering for PHSI can be taken as a multiview clustering task. There have been many attempts to extend

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the subspace models to multiview application scenarios, such as multimedia and multilingual documents [26]–[28]. In the past few years, several studies have found that integrating multiview data into a tensor can help seek the joint latent subspace structure. By learning projections from the all the modes of a third-order tensor, not only the row and column spaces of representation matrices but also the relationships among different representation matrices can be exploited. Zhang *et al.* [29] introduced the low-rank tensor constrained multiview subspace clustering (LT-MSC) model regarding the subspace representation matrices of different views as a tensor. Yin *et al.* [30] adopted the tensor nuclear norm (TNN) and proposed the subspace clustering for multiview data in the third-order tensor space (SCMV-3DT) algorithm, aiming to learn the view-specific representation as well as the low-rank tensor representation. Xiao *et al.* [31] and Xie *et al.* [32] modified the TNN using the tensor-singular value decomposition (t-SVD) and achieved the state-of-the-art multiview clustering performance. Very recently, Xie *et al.* [33] designed a weighted TNN minimization (WTNNM) that assigned different weights to singular values adaptively. Considering data features reside in multiple nonlinear subspaces instead of linear ones, Xie *et al.* [34] proposed the hyper-Laplacian regularized multilinear multiview self-representation model (HLR-M2VS).

Since a tensor is an efficient representation of multidimensional data, it is natural to analyze HSI data from the tensor perspective. HSIs consisting of thousands of pixels usually contain just a few dozen species, and the spectra corresponding to the same species are similar in the subspace expression. At the same time, the hypothesis that adjacent pixels in HSIs have a high probability of being of the same species is widely adopted in HSI studies. Thus, HSIs are considered to show significant low-rank characteristics in both the spectral and spatial domains. Recently, the tensor-based subspace representation has been widely applied to HSI processing tasks. Dian and Li [35] and Xue *et al.* [36] introduced the tensor representation to make use of high spectral correlations in HSI super-resolution research. Xue *et al.* [37] and Zhang *et al.* [38] also utilized the tensor decomposition to find the nonlocal low-rank features in HSI denoising research. Fejjari *et al.* [39] used the tensor representation and the intrinsic decomposition (ID) to integrate spectral and spatial information in HSI classification research. Furthermore, Chen *et al.* [40] combined the CANDECOMP PARAFAC decomposition (CPD) of tensor representation and the convolutional neural network (CNN) and achieved high HSI classification accuracy. Huang *et al.* [41] successfully achieved superior performance in HSI clustering tasks by converting HSI into higher-dimensional data using local binary pattern (LBP) and Gabor filtering and then representing the data with a tensorial form. However, because of the high computational complexity, subspace-based methods require large amounts of computational memory and large running time consumption. Both HSI and PHSI are usually associated with large image scales and high dimensionality, so the computational complexity becomes one of the major limitations of subspace-based approaches in practical applications.

In this article, the tensorial polarimetric-spectral multiview subspace clustering (TPS-MSC) algorithm is proposed. It combines the multiview polarimetric and spectral properties of targets, performing better than in the case where only one view is considered. To address the computational complexity limitation, by pre-clustering each view independently, TPS-MSC selects a part of the data from each view instead of all the data as a dictionary for the representation. Benefiting from the low-rank structure among the representation matrices of different views, the improved TNN (ITNN) [42] is introduced to take full advantage of interview properties. Then, the model is modified based on the consistency exhibited among all individual views during the pre-clustering step. The architecture of the proposed TPS-SSC algorithm is as shown in Fig. 1. Also, the main contributions of this article are summarized as follows.

1) For the first time, the clustering of PHSI is specified as a multiview clustering problem. An efficient multiview subspace clustering model for PHSI with tensorial low-rank constraint and multiview pre-clustering consistency constraint is established.
2) An effective approach to optimize the proposed TPS-MSC model is designed via the alternative direction minimization of multipliers (ADMM) algorithm [34], [35].
3) Comprehensive experiments on two PHSI datasets are carried out, and the experimental results demonstrate the better clustering performance of the TPS-MSC algorithm compared with the existing methods.

The rest of this article is structured as follows. Section II reviews the typical multiview subspace clustering model and introduces ITNN briefly. Section III presents the TPS-MSC algorithm in detail. Section IV gives the experimental results. Section V concludes this article.

II. Notations and Preliminaries

A. Description of PHSIs

PHSI is usually represented by images with four different polarization directions under different spectral bands (for example, $I_{0^\circ}(\lambda)$, $I_{45^\circ}(\lambda)$, $I_{90^\circ}(\lambda)$, and $I_{135^\circ}(\lambda)$, where λ is the wavelength). Also, PHSI can be described by the Stokes vector $S(\lambda)$, which is composed of four Stokes parameters

$$S(\lambda) = \begin{bmatrix} S_0(\lambda) \\ S_1(\lambda) \\ S_2(\lambda) \\ S_3(\lambda) \end{bmatrix} = \begin{bmatrix} I_{0^\circ}(\lambda) + I_{90^\circ}(\lambda) \\ I_{0^\circ}(\lambda) - I_{90^\circ}(\lambda) \\ I_{45^\circ}(\lambda) - I_{135^\circ}(\lambda) \\ I_R(\lambda) - I_L(\lambda) \end{bmatrix} \tag{1}$$

where I_R and I_L denote the right-handed circular polarization state and the left-handed circular polarization state, respectively. Under natural conditions, $S_3(\lambda)$ is usually negligible.

Then, the degree of linear polarization (DOLP) and the angle of polarization (AOP) can be calculated by

$$\mathrm{DOLP}(\lambda) = \frac{\sqrt{S_1(\lambda)^2 + S_2(\lambda)^2}}{S_0(\lambda)} \tag{2}$$

$$\mathrm{AOP}(\lambda) = \arctan\left(\frac{S_2(\lambda)}{S_1(\lambda)}\right). \tag{3}$$

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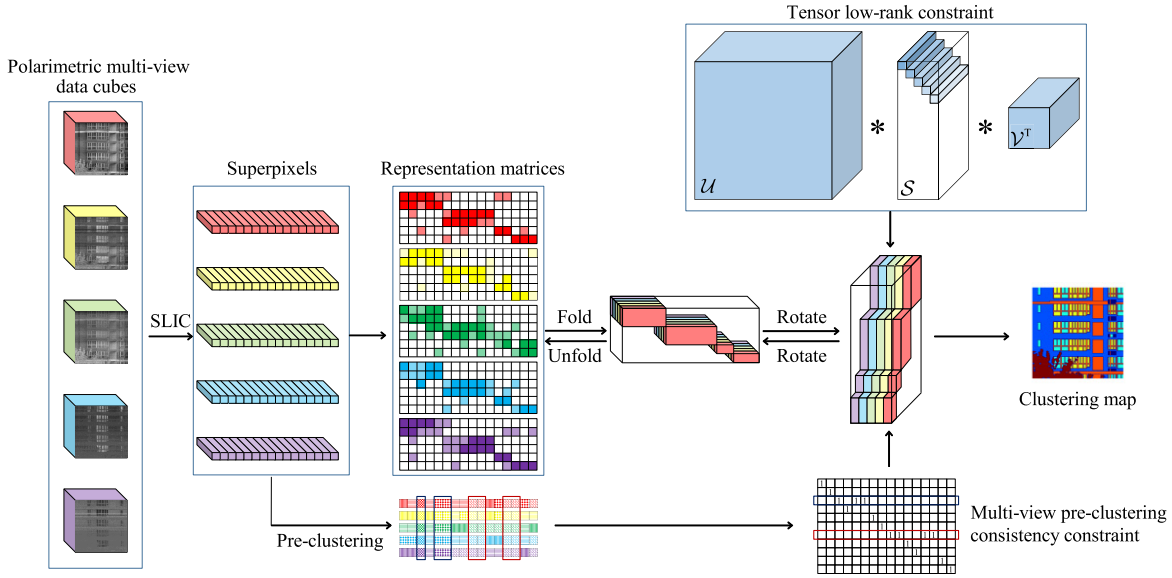


Fig. 1. Architecture of the proposed TPS-MSC algorithm.

In this article, the PHSI datasets consist of five components (i.e., five views): $S_0(\lambda)$, $S_1(\lambda)$, $S_2(\lambda)$, DOLP(λ), and AOP(λ).

B. Multiview Subspace Clustering

Suppose the given N data points $\{X^{(v)}\}_{v=1}^{V}$ have V views, where $X^{(v)} = [x_1^{(v)}, x_2^{(v)}, \ldots, x_N^{(v)}] \in \mathbb{R}^{d_v \times N}$ denotes the vth view with d_v dimensions. For an individual view, under the assumption that every data point in a subspace can be represented as the linear combination of just a few other points in that same subspace, $X^{(v)}$ itself is utilized as the dictionary to build the subspace representation model as

$$X^{(v)} = X^{(v)}C^{(v)} + E^{(v)} \tag{4}$$

where $C^{(v)}$ representation denotes the representation coefficient matrix, and $E^{(v)}$ denotes the representation error.

The existing multiview subspace clustering algorithms usually differ in the representation model, but the general form of them can be summarized as

$$\min_{C^{(v)}} \sum_{v=1}^{V} \Theta\left(X^{(v)}, X^{(v)}C^{(v)}\right) + \lambda\Omega\left(C^{(v)}\right)$$
$$\text{s.t. } X^{(v)} = X^{(v)}C^{(v)} + E^{(v)} \quad \forall v \tag{5}$$

where $\Theta(\cdot,\cdot)$ denotes the loss function, $\Omega(\cdot)$ denotes the regularized term, and λ denotes the trade-off parameter balancing the regularized term and loss.

The representation matrices $\{C^{(v)}\}_{v=1}^{V}$, which express each data as a linear combination of other data, measure the similarities between samples. Then, a reliable clustering result can be obtained with $\{C^{(v)}\}_{v=1}^{V}$ by applying a certain basis clustering method, such as k-means and spectral clustering.

C. Improved TNN

A multiview subspace model that handles each view independently does not take full advantage of interview properties. Recently, some methods based on tensorial norms have been verified to be effective in exploring the low-rank structure among different views. Here, we briefly introduce a representative one of them, the ITNN [42].

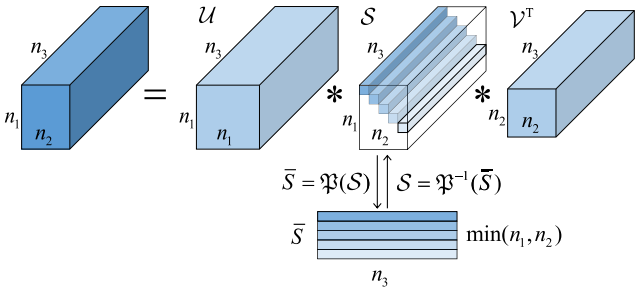


Fig. 2. Illustration of t-SVD for an $n_1 \times n_2 \times n_3$ tensor and transforms between core tensor $\mathcal{S}$ and core matrix $\bar{S}$.

Algorithm 1 t-SVD

Input: tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$.
1. $\hat{A} = \text{fft}(\mathcal{A}, [], 3)$.
2. **for** $i = 1, 2, \ldots, n_3$ **do**
   $[U, S, V^T] = \text{SVD}(\hat{A}^{(i)})$.
   $\hat{\mathcal{U}}^{(i)} = U$, $\hat{\mathcal{S}}^{(i)} = S$, $\hat{\mathcal{V}}^{(i)} = V$.
   end
3. $\mathcal{U} = \text{ifft}(\hat{\mathcal{U}}, [], 3)$, $\mathcal{S} = \text{ifft}(\hat{\mathcal{S}}, [], 3)$, $\mathcal{V} = \text{ifft}(\hat{\mathcal{V}}, [], 3)$.
4. **Output:** $\mathcal{U}$, $\mathcal{S}$, $\mathcal{V}$.

As shown in Fig. 2, a third-order tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ can be factorized by the t-SVD technology as

$$\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T \tag{6}$$

where $\mathcal{U} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ and $\mathcal{V}^T \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal tensors, and the core tensor $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an f-diagonal tensor. The calculation process of t-SVD is summarized as in Algorithm 1.

The core tensor $\mathcal{S}$ can be converted to a 2-D core matrix $\bar{S}$ whose entries come from the diagonal elements of $\mathcal{S}$, as shown

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in Fig. 2 (lower part). The conversion operators are written as $\bar{S} = \mathfrak{P}(\mathcal{S})$ and $\mathcal{S} = \mathfrak{P}^{-1}(\bar{S})$.

The core tensor $\mathcal{S}$ can be converted to a 2-D core matrix $\bar{S}$ whose entries come from the diagonal elements of $\mathcal{S}$. Here, the conversion operator and the corresponding inverse operation are written as $\bar{S} = \mathfrak{P}(\mathcal{S})$ and $\mathcal{S} = \mathfrak{P}^{-1}(\bar{S})$. The core tensor $\mathcal{S}$ is an f-diagonal tensor, and only the values of $\mathcal{S}(i, i, :), i = 1, 2, \ldots, \min(n_1, n_2)$ are non-zero (i.e., the blue part of core tensor $\mathcal{S}$ in Fig. 2). The operator $\mathfrak{P}(\mathcal{S})$ reshapes $\mathcal{S}(i, i, :), i = 1, 2, \ldots, \min(n_1, n_2)$ to a matrix $\bar{S}$ with size $\min(n_1, n_2) \times n_3$, as shown in Fig. 2 (lower part).

Liu *et al.* found that the low rank structure still exists in $\bar{S}$ and proposed the ITNN, which can further extract low rank components for the core matrix. The definition of ITNN is

$$||\mathcal{A}||_{\text{ITNN}} = ||\bar{S}||_* + \gamma\, ||\mathcal{U} * \mathcal{S} * \mathcal{V}^T||_* \tag{7}$$

where $\gamma = 1/(\max(n_1, n_2)n_3)^{1/2}$ is a parameter to balance the two terms.

III. Methodology

A. Subspace Representation With a Small Size Dictionary

The traditional subspace clustering models use all data points as a dictionary, which leads to high computational complexity in the optimization process. Superpixel segmentation of images can reduce the complexity in time and in space to some extent. However, subspace clustering approaches are still very demanding on computational memory, even if the original images have been partitioned into thousands of superpixels. Some research constructed a structured dictionary that contains a small number of atoms, significantly improving the efficiency of subspace clustering [43]. Here, we also adopt several representative data instead of all data as a dictionary.

First, the original PHSIs can be segmented into N superpixels by the simple linear iterative clustering (SLIC) algorithm. The superpixel set will replace the pixels in the subsequent process as the target data to be clustered. In experiments, three bands close to the red-green-blue (RGB) channels of S_0-view are utilized to construct a false-color image for the SLIC algorithm. Also, the segmentation results for S_0-view will be directly applied to other views.

To select the representative superpixels, the k-means algorithm is applied to pre-clustering each view independently. In each view, superpixels are assigned into K clusters, where K is the number of species. The m samples closest to the cluster centers will be selected to construct the dictionary. Here, we denote the dictionary as $\{D^{(v)}\}_{v=1}^{V}$, where $D^{(v)} = [D_1^{(v)}, D_2^{(v)}, \ldots, D_M^{(v)}] \in \mathbb{R}^{d_v \times M}$, and $M = V \times K \times m$ is the total number of selected samples.

Then, the loss term in (5) becomes

$$\min_{E^{(v)}} \sum_{v=1}^{V} ||E^{(v)}||_{2,1} \quad \text{s.t. } X^{(v)} = D^{(v)}C^{(v)} + E^{(v)}. \tag{8}$$

The amount of information contained in each view of PHSI is different. Usually, S_0-view can characterize most features of targets, and other views do not have the ability to distinguish all species individually. Therefore, the similarity of S_0-view needs to be considered when calculating the representation coefficient matrix $C^{(v)}$ of the vth-view. That means, for the ith and the jth superpixels, if they are similar in S_0-view, the corresponding vectors $C_i^{(v)}$ and $C_j^{(v)}$ (i.e., the ith column and the jth column of $C^{(v)}$) are desired to be similar. In light of this, the optimization problem becomes

$$\begin{aligned} &\min_{E^{(v)}} \sum_{v=1}^{V} \left(||E^{(v)}||_{2,1} + \lambda_1 \sum_{i=1}^{N}\sum_{j=1}^{N} \left\| C_i^{(v)} - C_j^{(v)} \right\|_2^2 W_{ij}^{(1)} \right) \\ &\text{s.t. } X^{(v)} = D^{(v)}C^{(v)} + E^{(v)} \quad \forall v \end{aligned} \tag{9}$$

where $W^{(1)}$ is the similarity matrix of the first view (i.e., S_0-view), and λ_1 is the regularization parameter. There are various methods to build the similarity matrix, such as the heat kernel weighting strategy and the 0–1 weighting strategy. Here, the heat kernel weighting strategy is adopted.

The optimization problem (9) is equivalent to

$$\begin{aligned} &\min_{E^{(v)}} \sum_{v=1}^{V} \left(||E^{(v)}||_{2,1} + \lambda_1 \text{Tr}\left(C^{(v)} L^{(1)} C^{(v)}\right) \right) \\ &\text{s.t. } X^{(v)} = D^{(v)}C^{(v)} + E^{(v)}, \quad v = 1, 2, \ldots, V \end{aligned} \tag{10}$$

where $L^{(1)} = \Delta^{(1)} - W^{(1)}$ is the Laplacian matrix of S_0-view, and $\Delta^{(1)}$ is a diagonal matrix with $\Delta_{ii}^{(1)} = \sum_{j=1}^{N} W_{ij}^{(1)}$.

B. Tensorial Low-Rank Constraint With ITNN

The representation matrices can reflect the similarity between the data and the dictionary. In vector $C_i^{(v)}$, the elements corresponding to the dictionary samples that are similar to the ith superpixel are greater. Since the same columns in $C^{(v)}$ of different views characterize the same superpixels, if we stack $\{C^{(v)}\}_{v=1}^{V}$ together into a tensor, as shown in Fig. 1 (middle part), significant low-rank features are desired to be observed.

Inspired by the benefits of ITNN, the tensor $\mathcal{C} \in \mathbb{R}^{M \times V \times N}$ is introduced to extract the low-rank structure among the view-independent representation matrices $\{C^{(v)}\}_{v=1}^{V}$. The tensor $\mathcal{C}$ is converted from $\{C^{(v)}\}_{v=1}^{V}$ by the operator $\mathcal{C} = \Phi(\{C^{(v)}\}_{v=1}^{V})$. This operator is a two-step process that first stacks all the 2-D representation matrices $\{C^{(v)}\}_{v=1}^{V}$ together to form a third-order tensor with size $M \times N \times V$, and then rotates the tensor to the size of $N \times V \times M$, since the t-SVD method performs SVD on each frontal slice of the tensor. Without rotating operation, SVD would be applied on the slice with size $M \times N$, and only the intraview low-rank property can be preserved. Here, the corresponding inverse operation is denoted as $\{C^{(v)}\}_{v=1}^{V} = \Phi^{-1}(\mathcal{C})$. Combining ITNN and (10) together, the optimization problem becomes

$$\begin{aligned} &\min_{E^{(v)}, C^{(v)}, \mathcal{Z}} \sum_{v=1}^{V} \left(||E^{(v)}||_{2,1} + \lambda_1 \text{Tr}\left(C^{(v)} L^{(1)} C^{(v)}\right) \right) + \lambda_2 ||\mathcal{Z}||_{\text{ITNN}} \\ &\text{s.t. } X^{(v)} = D^{(v)}C^{(v)} + E^{(v)}, \quad v = 1, 2, \ldots, V \\ &\mathcal{Z} = \mathcal{C}, \quad \mathcal{C} = \Phi\left(\left\{C^{(v)}\right\}_{v=1}^{V}\right) \end{aligned} \tag{11}$$

where λ_2 is a trade-off parameter.

Model (11) allows to learn the low-rank structure of the third-order tensor $\mathcal{C}$ as well as its core matrix. In a joint framework, both intraview and interview correlations are

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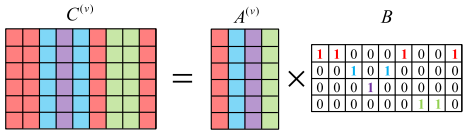


Fig. 3. Illustration of the multiview pre-clustering consistency.

extracted by dictionary-based representation and tensor-based representation.

C. Multiview Pre-Clustering Consistency Constraint

When the dictionary is constructed through pre-clustering, the respective clustering results for each view are also generated. The pre-clustering results of various views are usually different, but there might be some superpixels, which are always assigned into the same clusters. Superpixels assigned to the same cluster in all views have a high probability of belonging to the same species. For example, for the two PHSI datasets adopted in the experiments of this article, the pre-clustering results of these superpixels achieve the high accuracies of over 85% and over 95%, respectively.

The small amount of error that still exists, as well as the unevenness of the data distribution, prevents these data from being used directly as training samples to classify other data. But, there is still a way to retain this multiview pre-clustering consistency in the steps of representing learning.

Intuitively, data showing consistency in pre-clustering should share the same representation coefficient vector. Therefore, we decompose the representation coefficient matrix as $C^{(v)} = A^{(v)}B$, where B is generated based on the results of pre-clustering of all views. In matrix B, each row represents data that are per-clustered into the same class in all view.

In each row of matrix B, the elements corresponding to data sharing the same representation coefficient vector are set to 1, while the other elements are set to 0. If a superpixel does not exhibit consistency among views, it can be considered to share a representation coefficient vector only with itself. The illustration of the multiview pre-clustering consistency is as shown in Fig. 3.

With the multiview pre-clustering consistency, the overall version of our model becomes

$$\begin{aligned}\min_{E^{(v)},C^{(v)},\mathcal{Z}} &\sum_{v=1}^{V}\left(||E^{(v)}||_{2,1}+\lambda_1\mathrm{Tr}\left(C^{(v)}L^{(1)}C^{(v)}\right)\right)+\lambda_2||\mathcal{Z}||_{\mathrm{ITNN}}\\ \text{s.t. } & X^{(v)} = D^{(v)}C^{(v)} + E^{(v)}, \quad C^{(v)} = A^{(v)}B\\ & v = 1, 2, \ldots, V\\ & \mathcal{Z} = \mathcal{C}, \quad \mathcal{C} = \Phi\left(\left\{C^{(v)}\right\}_{v=1}^{V}\right).\end{aligned} \tag{12}$$

D. Optimization

The ADMM algorithm [44], [45] is used to solve model (12). The augmented Lagrangian function of (12) is

$$\begin{aligned}&\mathcal{L}_{\mu>0}\left(\left\{E^{(v)}; C^{(v)}\right\}_{v=1}^{V}, \mathcal{Z}\right)\\ &= \sum_{v=1}^{V}\left(||E^{(v)}||_{2,1} + \lambda_1\mathrm{Tr}\left(C^{(v)}L^{(1)}C^{(v)}\right)\right) + \lambda_2||\mathcal{Z}||_{\mathrm{ITNN}}\\ &+ \sum_{v=1}^{V}\Big(< Y_1^{(v)}, X^{(v)} - D^{(v)}C^{(v)}\\ &\qquad - E^{(v)} > + \frac{\mu}{2}||X^{(v)} - D^{(v)}C^{(v)} - E^{(v)}||_2^2\\ &\qquad + < Y_2^{(v)}, C^{(v)} - A^{(v)}B > + \frac{\mu}{2}||C^{(v)} - A^{(v)}B||_2^2\Big)\\ &+ < \mathcal{Y}_3, \mathcal{C} - \mathcal{Z} > + \frac{\mu}{2}||\mathcal{C} - \mathcal{Z}||_2^2\end{aligned} \tag{13}$$

where $Y_1^{(v)}$, $Y_2^{(v)}$, and $\mathcal{Y}_3$ are the Lagrange multipliers, μ is the penalty factor, $< \cdot, \cdot >$ denotes the inner product, and $\{Z^{(v)}\}_{v=1}^{V} = \Phi^{-1}(\mathcal{Z})$.

Based on the ADMM framework, (13) can be minimized by updating one variable with other variables fixed iteratively. The updating process for each iteration is as follows.

1) Updating $\{C^{(v)}\}_{v=1}^{V}$*:* When other variables are fixed except for $\{C^{(v)}\}_{v=1}^{V}$, the optimization problem (13) becomes

$$\begin{aligned}\min_{C^{(v)}} \sum_{v=1}^{V}\Bigg(&\lambda_1\mathrm{Tr}\left(C^{(v)}L^{(1)}C^{(v)}\right)\\ &+ \frac{\mu}{2}\left\|X^{(v)} - D^{(v)}C^{(v)} - E^{(v)} + \frac{Y_1^{(v)}}{\mu}\right\|_2^2\\ &+ \frac{\mu}{2}\left\|C^{(v)} - A^{(v)}B + \frac{Y_2^{(v)}}{\mu}\right\|_2^2\\ &+ \frac{\mu}{2}\left\|C^{(v)} - Z^{(v)} + \frac{Y_3^{(v)}}{\mu}\right\|_2^2\Bigg)\end{aligned} \tag{14}$$

where $\{Y_3^{(v)}\}_{v=1}^{V} = \Phi^{-1}(\mathcal{Y}_3)$. For each view, $C^{(v)}$ can be updated independently by taking the derivative with respect to $C^{(v)}$ as zero

$$\begin{aligned}&\left(\mu D^{(v)T}D^{(v)} + 2\mu I\right)C^{(v)} + C^{(v)} \cdot 2\lambda_1 L^{(1)}\\ &= \mu D^{(v)T}\left(X^{(v)} - E^{(v)} + \frac{Y_1^{(v)}}{\mu}\right) + \mu\left(A^{(v)}B - \frac{Y_2^{(v)}}{\mu}\right)\\ &\quad + \mu\left(Z^{(v)} - \frac{Y_3^{(v)}}{\mu}\right).\end{aligned} \tag{15}$$

Equation (15) can be written as a typical Sylvester equation $a \cdot C^{(v)} + C^{(v)} \cdot b = c$ with $a = (\mu D^{(v)T}D^{(v)} + 2\mu I)$, $b = 2\lambda_1 L^{(1)}$, and $c = \mu D^{(v)T}(X^{(v)} - E^{(v)} + Y_1^{(v)}/\mu) + \mu(A^{(v)}B - Y_2^{(v)}/\mu) + \mu(Z^{(v)} - Y_3^{(v)}/\mu)$.

The Sylvester equation can be solved by the Bartels–Stewart algorithm. In experiments, the function "sylvester" of the MATLAB Control Toolbox is used to solve this problem.

2) Updating $\{A^{(v)}\}_{v=1}^{V}$*:* When other variables are fixed except for $\{A^{(v)}\}_{v=1}^{V}$, the optimization problem (13) becomes

$$\min_{A^{(v)}} \sum_{v=1}^{V}\frac{\mu}{2}\left\|C^{(v)} - A^{(v)}B + \frac{Y_2^{(v)}}{\mu}\right\|_2^2. \tag{16}$$

Similar to $C^{(v)}$, $A^{(v)}$ of each view can be updated independently by taking the derivative with respect to $A^{(v)}$ as zero

$$A^{(v)}B^TB = \left(C^{(v)} + \frac{Y_2^{(v)}}{\mu}\right)B. \tag{17}$$

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Equation (15) can be regarded as a Sylvester equation with $a = 0$, $b = B^T B$, and $c = (C^{(v)} + Y_2^{(v)}/\mu)B$. Here, it can also be solved by the function "sylvester."

3) Updating $\{E^{(v)}\}_{v=1}^V$*:* When other variables are fixed except for $\{E^{(v)}\}_{v=1}^V$, the optimization problem (13) becomes

$$\min_{E^{(v)}} \sum_{v=1}^{V} \left(||E^{(v)}||_{2,1} + \frac{\mu}{2} \left\| X^{(v)} - D^{(v)}C^{(v)} - E^{(v)} + \frac{Y_1^{(v)}}{\mu} \right\|_2^2 \right). \tag{18}$$

Referring to the work of Liu *et al.* [46], the solution of $E^{(v)}$ can be obtained by

$$E_{:,j}^{(v)} = \begin{cases} \dfrac{\left\| H_{:,j}^{(v)} \right\|_2 - \frac{1}{\mu}}{\left\| H_{:,j}^{(v)} \right\|_2} H_{:,j}^{(v)}, & \text{if } \left\| H_{:,j}^{(v)} \right\|_2 > \dfrac{1}{\mu} \\ 0, & \text{otherwise} \end{cases} \tag{19}$$

where $H^{(v)} = X^{(v)} - D^{(v)}C^{(v)} + Y_1^{(v)}/\mu$, and $E_{:,j}^{(v)}$ and $H_{:,j}^{(v)}$ denote the jth column of $E^{(v)}$ and $H^{(v)}$, respectively.

4) Updating $\mathcal{Z}$*:* When other variables are fixed except for $\mathcal{Z}$, the optimization problem (13) becomes

$$\min_{\mathcal{Z}} \lambda_2 ||\mathcal{Z}||_{\text{ITNN}} + \frac{\mu}{2} \left\| \mathcal{Z} - \mathcal{C} + \frac{\mathcal{Y}_3}{\mu} \right\|_F^2. \tag{20}$$

Referring to [42], (20) can be solved into two steps: the first to minimize the core matrix, and the second to minimize the TNN. First, the core matrix minimization can be written as

$$\min_{\mathfrak{P}(\mathcal{S})} \lambda_2 ||\mathfrak{P}(\mathcal{S})||_* + \frac{1}{2\gamma} \left\| \mathcal{M} - \mathcal{C} + \frac{\mathcal{Y}_3}{\mu} \right\|_F^2 \tag{21}$$

where $\gamma = 1/(\max(M, V)N)^{1/2}$ is a regularization parameter, $\mathcal{S}$ comes from the t-SVD of $(\mathcal{C} + \mathcal{Y}/\mu)$ as t-SVD$(\mathcal{C} + \mathcal{Y}/\mu) = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$, and $\mathcal{M}$ is a temporary variable.

By the point-wise soft thresholding operator $\text{sth}_\tau(X)$ and the point-wise singular value thresholding operator $\text{svt}_\tau(X)$, which are, respectively, defined as $\text{sth}_\tau(X) = sgn(X) \cdot \max(|X| - \tau, 0)$ and $\text{svt}_\tau(X) = U\text{sth}_\tau(\Sigma)V^T$, where U, Σ, and V come from the SVD of matrix as $X = U\Sigma V^T$, the core matrix $\bar{S}$ can be solved by

$$\bar{S} = \text{svt}_\gamma(\mathfrak{P}(\mathcal{S})). \tag{22}$$

With the low rank core matrix, a tensor $\mathcal{T}$ can be re-contracted by

$$\mathcal{T} = \mathcal{U} * \mathfrak{P}^{-1}(\bar{S}) * \mathcal{V}^T. \tag{23}$$

Then, the second step minimizing the TNN is as

$$\min_{\mathcal{Z}} ||\mathcal{Z}||_* + \frac{\mu}{4\lambda_2} ||\mathcal{Z} - \mathcal{T}||_F^2. \tag{24}$$

The optimization problem (24) can be solved by

$$\mathcal{Z} = \text{tsvt}_{2\lambda_2/\mu}(\mathcal{T}) \tag{25}$$

where $\text{tsvt}_\tau(\mathcal{X})$ denotes the tensor singular value thresholding operator, which uses the $\text{sth}_\tau(X)$ operator for matrices $\hat{\mathcal{S}}(:, :, i), \forall i$, where $\hat{\mathcal{S}}$ is the Fourier transform of $\mathcal{S}$, and $\mathcal{S}$ comes from the t-SVD of $\mathcal{X}$ [42].

5) Updating $\{Y_1^{(v)}\}_{v=1}^V$*,* $\{Y_2^{(v)}\}_{v=1}^V$*, and* $\mathcal{Y}_3$*:* The Lagrange multipliers $\{Y_1^{(v)}\}_{v=1}^V$, $\{Y_2^{(v)}\}_{v=1}^V$, and $\mathcal{Y}_3$ are, respectively, updated by

$$Y_1^{(v)} = Y_1^{(v)} + \mu\left(X^{(v)} - D^{(v)}C^{(v)} - E^{(v)}\right) \tag{26}$$

$$Y_2^{(v)} = Y_2^{(v)} + \mu\left(C^{(v)} - A^{(v)}B\right) \tag{27}$$

$$\mathcal{Y}_3 = \mathcal{Y}_3 + \mu(\mathcal{C} - \mathcal{Z}). \tag{28}$$

The updating will continue until it meets the maximum number of iterations or the stopping criterion, which is given by

$$\max \begin{pmatrix} \Sigma_{v=1}^V ||X^{(v)} - D^{(v)}C^{(v)} - E^{(v)}||_\infty \\ \Sigma_{v=1}^V ||C^{(v)} - A^{(v)}B||_\infty \\ \Sigma_{v=1}^V ||C^{(v)} - Z^{(v)}||_\infty \end{pmatrix} < \varepsilon \tag{29}$$

where ε is a small value, which is set as 10^{-5} in our experiments.

After the updating iterations are completed, due to the particularity of S_0-view and DOLP-view, which will be analyzed in Section IV-C, the final representation matrix is constructed as

$$C = C^{(1)} \cdot W^{(4)} + \sum_{v=2}^{V} \left(C^{(v)} \cdot W^{(1)}\right). \tag{30}$$

The weighted matrices $W^{(1)} \in \mathbb{R}^{M \times N}$ and $W^{(4)} \in \mathbb{R}^{M \times N}$ are defined as

$$W^{(1)}(i, j) = \frac{1}{||D^{(1)}(i) - X^{(1)}(j)||_2^2 + \delta} \tag{31}$$

$$W^{(4)}(i, j) = \frac{1}{||D^{(4)}(i) - X^{(4)}(j)||_2^2 + \delta} \tag{32}$$

where δ is a very small value preventing the denominator from being 0, $D^{(1)}(i)$ and $D^{(4)}(i)$ denote the vectors of the ith sample in the dictionary in S_0-view and DOLP-view, and $X^{(1)}(i)$ and $D^{(4)}(i)$ denote the vectors of the jth superpixel in S_0-view and DOLP-view, respectively.

Then, a traditional clustering algorithm, such as k-means or spectral clustering, can be applied on the constructed affine matrix C to obtain the final clustering results of all superpixels.

The proposed algorithm is summarized as Algorithm 2.

IV. Experimental Results and Discussion

To validate the effectiveness and the accuracy of the proposed TPS-SSC algorithm, a series of experiments are carried out on two datasets of PHSIs.

Section IV-A introduces the experimental settings, and Section IV-B displays the clustering performance of different methods. Sections IV-C and IV-D consider the effects of different multiview combinations and different tensorial norms, respectively. The analyses of parameters are in Section IV-E.

A. Experimental Settings

Two PHSI datasets are utilized to evaluate the performance of different methods. One of them, referred to as dataset A in this article, is measured by the CIISP, which can simultaneously acquire four Stokes vectors at 128 bands in

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Algorithm 2 TPS-MSC

Input:
Multi-view superpixel set $\{X^{(v)}\}_{v=1}^{V}$ and dictionary set $\{D^{(v)}\}_{v=1}^{V}$;
parameters λ_1, λ_2 and m;
the number of desired clusters K.
Initialization:
$C^{(v)} = 0, A^{(v)} = 0, E^{(v)} = 0, Z^{(v)} = 0, \forall v$;
$Y_1^{(v)} = 0, Y_2^{(v)} = 0, Y_3^{(v)} = 0, \forall v$;
$\mu = 1$, $\rho = 1.4$, $\rho_{\max} = 10^6$.
while *not converged* **do**
  for $v = 1, 2, \ldots, V$ **do**
    Update $C^{(v)}$ via the Bartels–Stewart algorithm to solve the Sylvester equation (15).
    Update $A^{(v)}$ via the Bartels–Stewart algorithm to solve the Sylvester equation (17).
    Update $E^{(v)}$ via (19).
  end
  Update $\mathcal{Z}$ via (22) and (25).
  for $v = 1, 2, \ldots, V$ **do**
    Update $Y_1^{(v)}$ and $Y_2^{(v)}$ via (26) and (27).
  end
  Update $\mathcal{Y}_3$ via (28)
  Update μ via $\mu = \min(\mu_{\max}, \mu\rho)$.
end
Compute C via (30).
Apply the k-means algorithm on C.
Output:
Clustering results of all superpixels.

480–960 nm [6]. CIISP modulates the polarization state onto several channels of the interferogram, and the Stokes vectors of targets can be obtained by Fourier transforming the different channels of the interferogram. This instrument is designed to be carried on a satellite for remote sensing detection, but it is still in the stage of ground verification. We used it to detect urban targets at about 2 km away, and the obtained PHSIs have a size of 512 × 512 pixels. Another dataset, referred to as dataset B in this article, was collected by Zhao *et al.* [47] of Northwestern Polytechnical University and covers 33 spectral bands in 400–720 nm. They conducted a collection of outdoor experiments using a liquid-crystal tunable filter (LCTF)-based imaging spectropolarimeter to obtain PHSI datasets for target detection and classification, which can be downloaded from the website http://jszy.nwpu.edu.cn/zhaoyq.html. The image size is 1360 × 1036 pixels, of which a 512 × 512 area is captured in this article.

Fig. 4(a)–(e) and (g)–(k) shows the S_0, S_1, S_2, DOLP, and AOP images at 690 and 778 nm of dataset A, respectively. Fig. 4(m)–(q) and (s)–(w) shows the S_0, S_1, S_2, DOLP, and AOP images at 480 and 690 nm of dataset B, respectively. A point worth making is that the measured S_0 is what is commonly referred to as radiance, and it contains most of the information needed to distinguish the target.

The two datasets have five and six species to be considered, respectively, and the labeled categories as well as the corresponding number of pixels in ground truth (GT) are shown in Table I. Polarization varies significantly from the roughness of the surface, and urban targets usually have a smooth surface. Dataset A contains several man-made targets that can reflect the ability of spectra and polarization to discriminate between different artifacts. Dataset B, on the other hand, is primarily concerned with the distinction between artifacts and non-artifacts.

Before applying clustering algorithms, images will be partitioned into thousands of superpixels by the SLIC algorithm. If there are too few superpixels, a single superpixel may have more than one category, resulting in poor clustering performance. But, if there are too many superpixels, the memory required and the time consumed increase dramatically. In our experiments, the Canny operator [48] is applied to transfer the false color image to the edge image, and then, the number of superpixels is determined by

$$P = P_0 \times \frac{N_{\text{non-zero}}}{N_{\text{all}}} \tag{33}$$

where $N_{\text{non-zero}}$ denotes the number of non-zero values in the edge image, N_{all} denotes the number of all pixels, and P_0 denotes the basis number of superpixels, which is empirically defined as 20 000 in the experiments. Fig. 4(l) and (x) shows the superpixel segmentation results.

In the dictionary constructing process, four views of S_0, S_1, S_2, and DOLP are selected to generate the dictionary, and the value of m is set as 5. Therefore, the numbers of atoms in dictionaries of dataset A (five clusters) and dataset B (six clusters) are 100 and 120, respectively.

To examine the performance of TPS-MSC, we compare it with four single-view clustering methods: k-means, rank-two nonnegative matrix factorization (R2NMF) [49], SSC [19], and low-rank SSC (LRSSC) [50], and five multiview clustering methods: multiview LRSSC (MLRSSC) [51], SCMV-3DT [30], low-rank preserving t-linear projection (LRP-tP) [31], WTNNM [33], and multilayer sparsity-based tensor decomposition (MLSTD) [52]. For single-view clustering methods, the view with the best performance, i.e., the S_0-view, is adopted as input.

The quantitative metrics to measure the clustering performance are the overall accuracy (OA), the kappa coefficient (κ), the average accuracy (AA), and running time.

B. Clustering Results and Discussion

The clustering results obtained with different methods of the two datasets are shown in Figs. 5 and 6, respectively. Figs. 5(j) and 6(j) indicate the GT in which some blurred regions corresponding to complex features are unlabeled, due to the insufficient spatial resolution of detection instruments. The quantitative evaluation of clustering results can be found in Tables II and III.

As can be seen from Tables II and III, two single-view subspace-based methods SSC and LRSSC perform better than k-means and R2NMF in OA, while they need much more running time. In Fig. 4(a) and (g), the S_0-view fails to clearly distinguish between windows with curtains and the surrounding blue-gray painted wall. Therefore, for dataset A,

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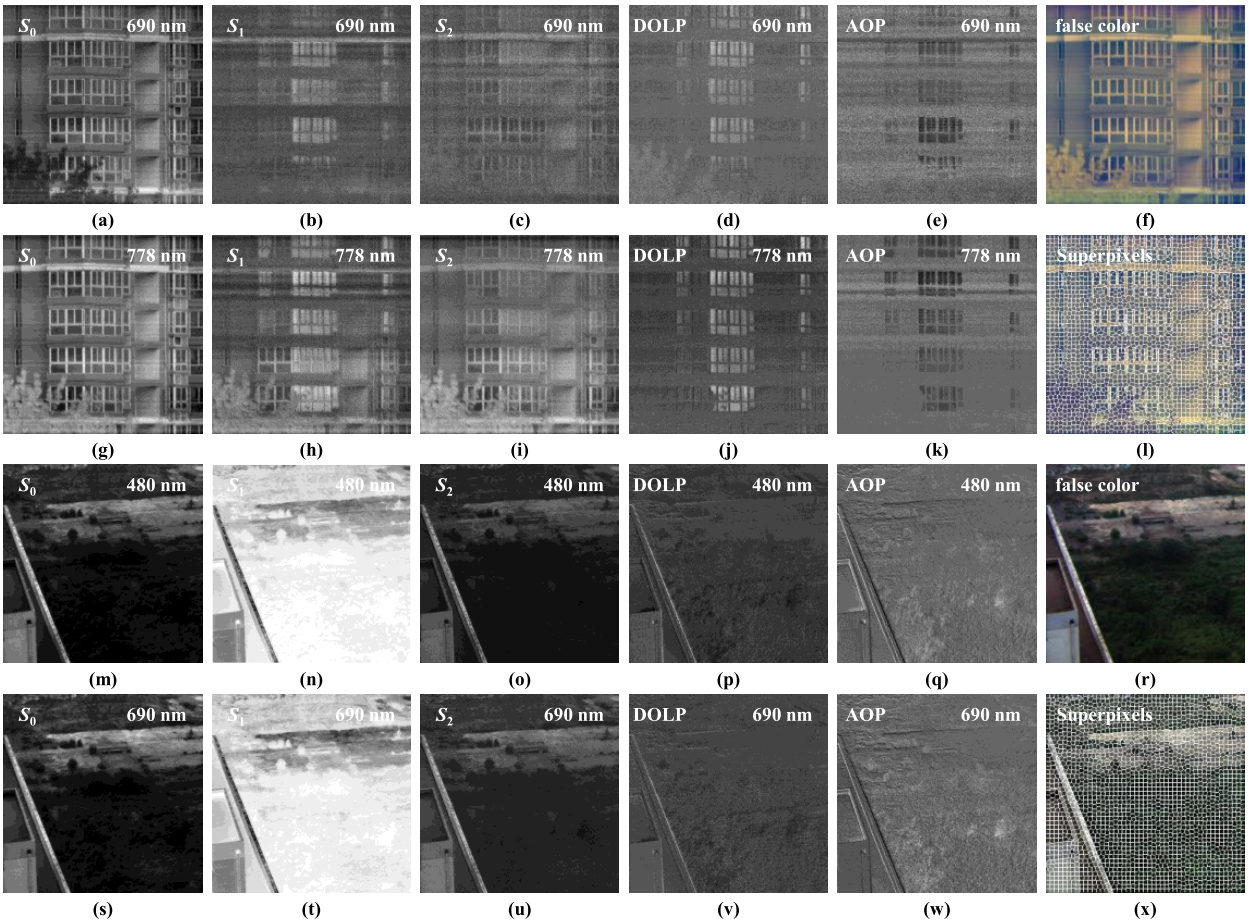


Fig. 4. Images of two datasets. (a)–(e) S_0, S_1, S_2, DOLP, and AOP images at 690 nm of dataset A. (f) False color image of dataset A. (g)–(k) S_0, S_1, S_2, DOLP, and AOP images at 778 nm of dataset A. (l) Superpixel segmentation of dataset A. (m)–(q) S_0, S_1, S_2, DOLP, and AOP images at 480 nm of dataset B. (r) False color image of dataset B. (s)–(w) S_0, S_1, S_2, DOLP, and AOP images at 690 nm of dataset B. (x) Superpixel segmentation of dataset B.

TABLE I
REFERENCE CLASSES WITH NUMBER OF SAMPLES FOR TWO DATASETS

Dataset A (512*512 pixels)			Dataset B (512*512 pixels)		
Class	Cover species	Number of samples	Class	Cover species	Number of samples
1	Blue-gray painted wall	90070	1	White wall	12023
2	Windows	22976	2	Blue wall	3676
3	Windows with curtains	26786	3	Black smooth ground	14292
4	White wall & window edges	54996	4	Vegetation	125803
5	Leaves	23703	5	White smooth ground	34706
6	\	\	6	Rough ground	33783
Total	218531		Total	224283	

all four single-view methods show low accuracy on windows with curtains, which directly leads to low OAs. The best performed single-view method LRSSC achieves 66.13% and 85.19% of OA on dataset A and dataset B, but it runs for more than 220 s.

Considering the multiview feature, the OA increases substantially. As a comparison, MLRSSC achieves the OA improvements of 7.91% and 2.7% over LRSSC on dataset A and dataset B, respectively. The tensorial models SCMV-3DT, LRP-tP, WTNNM, and MLSTD, which achieve the OAs of over 75% and 88% on two datasets, further improve the clustering performance compared with MLRSSC. This demonstrates that tensorizing the representation matrices does allow for better learning of low-rank features among different views. However, the increasing number of views means a larger amount of data, which becomes a heavy burden on computation.

The proposed TPS-MSC algorithm enables high OA with short time consumption. It only takes about 91.02 and 77.68 s on two datasets, much shorter than the over 750 s of running time acquired for the five multiview methods using all the data as a dictionary. With the ITNN and the multiview

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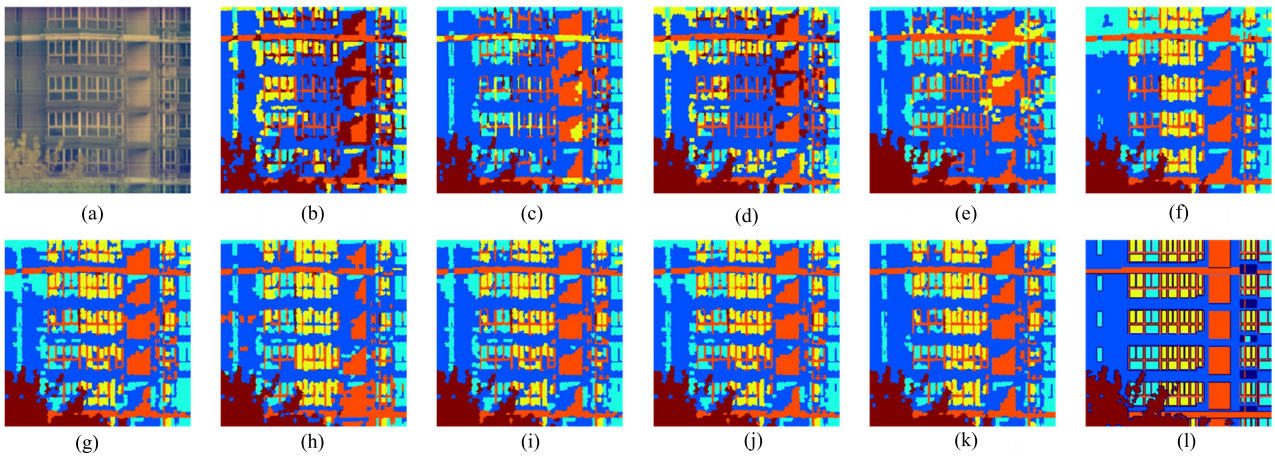


Fig. 5. Cluster maps of the different methods of dataset A. (a) False color. (b) k-means with S_0-view. (c) R2NMF with S_0-view. (d) SSC with S_0-view. (e) LRSSC with S_0-view. (f) MLRSSC. (g) SCMV-3DT. (h) LRP-tP. (i) WTNNM. (j) MLSTD. (k) TPS-MSC. (l) GT.

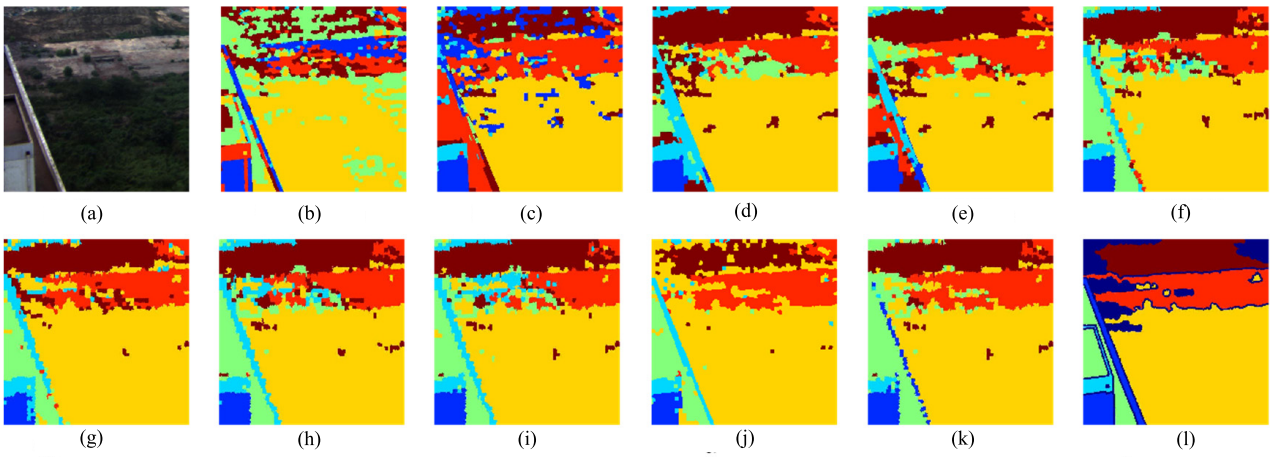


Fig. 6. Cluster maps of the different methods of dataset B. (a) False color. (b) k-means with S_0-view. (c) R2NMF with S_0-view. (d) SSC with S_0-view. (e) LRSSC with S_0-view. (f) MLRSSC. (g) SCMV-3DT. (h) LRP-tP. (i) WTNNM. (j) MLSTD. (k) TPS-MSC. (l) GT.

TABLE II
QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS OF DATASET A

Method	class	kmeans -S_0	R2NMF -S_0	SSC - S_0	LRSSC S_0	MLRSSC	SCMV -3DT	LRR -tP	WTNNM	MLSTD	TPS -MSC
Accuracy (%)	1	74.17	84.09	73.76	75.58	76.82	75.48	81.63	79.33	80.25	87.69
	2	57.53	76.85	81.14	56.08	59.03	80.48	76.94	79.86	80.43	71.85
	3	31.01	0.25	1.52	34.08	82.30	76.04	86.71	75.50	79.10	81.79
	4	44.11	49.85	65.84	54.59	64.64	65.82	59.39	65.09	66.04	67.19
	5	75.46	84.08	96.29	90.35	90.79	89.64	85.97	88.34	89.34	89.88
OA (%)		59.70	64.43	66.13	64.76	74.07	75.18	76.63	76.31	77.54	80.38
AA (%)		56.45	59.02	63.71	62.13	74.71	77.49	78.13	77.62	79.03	79.68
Kappa		0.452	0.498	0.532	0.515	0.646	0.663	0.678	0.676	0.693	0.727
Running time (s)		21.24	13.52	174.5	169.6	787.9	810.4	799.1	830.5	715.2	91.03

pre-clustering consistency constraint, TPS-MSC yields the best clustering performance. In particular, the OAs of our method on two datasets are as high as 80.38% and 92.17%, respectively.

C. Analysis of Multiview Combination

According to Tables II and III, multiview data allows for better separability of targets compared with single-view ones. But, it does not mean that more views are better. Table IV shows the clustering quantitative evaluation of TPS-MSC with various combinations of views, where view indexes 1–5 indicate the views of S_0, S_1, S_2, DOLP, and AOP, respectively. Since the other views are considered as supplements to S_0, S_0 will be retained in all combinations. Note that in two-view combinations, $W^{(4)}$ in (30) is modified as $W^{(v)}$, where v denotes the index

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TABLE III
QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS OF DATASET B

Method	class	kmeans -S_0	R2NMF -S_0	SSC - S_0	LRSSC S_0	MLRSSC	SCMV -3DT	LRR -tP	WTNNM	MLSTD	TPS -MSC
Accuracy (%)	1	75.69	69.64	60.83	66.25	58.07	58.07	48.08	57.26	51.94	65.57
	2	8.62	80.39	94.71	81.90	72.79	77.60	79.74	77.01	78.57	64.85
	3	66.22	0.84	32.56	7.87	96.50	95.06	92.04	96.97	91.15	98.56
	4	84.13	92.28	97.17	95.72	96.41	97.51	98.67	97.25	99.03	97.56
	5	36.63	88.55	60.98	78.83	62.71	61.10	74.59	67.76	85.84	83.29
	6	45.77	65.66	92.60	93.04	90.56	92.35	88.01	91.31	71.54	90.81
OA (%)		68.22	80.35	84.69	85.19	87.89	88.51	89.89	89.31	89.49	92.17
AA (%)		52.85	66.23	73.14	70.60	79.51	80.28	80.19	81.26	79.68	83.44
Kappa		0.531	0.697	0.760	0.767	0.811	0.820	0.838	0.833	0.827	0.876
Running time (s)		20.18	14.72	170.9	176.6	760.2	776.3	750.8	791.5	699.1	77.68

TABLE IV
OA AND RUNNING TIME WITH VARIOUS MULTIVIEW COMBINATIONS

Dataset A	1	1+2	1+3	1+4	1+5	1+2 +3+4	1+2+3 +4+5
OA (%)	65.19	77.85	67.62	79.51	74.25	80.38	78.22
Time (s)	51.63	71.97	68.52	71.25	70.56	91.03	111.3
Dataset B	1	1+2	1+3	1+4	1+5	1+2 +3+4	1+2+3 +4+5
OA (%)	84.99	87.73	88.96	91.82	79.28	92.17	89.37
Time (s)	42.50	53.45	54.26	58.22	55.54	77.68	100.5

View index: 1-S_0; 2-S_1; 3-S_2; 4-DOLP; 5-AOP.

corresponding to the other view in the combination apart from S_0.

For both datasets, the presence of the additional views improves the accuracy compared to clustering with S_0-view only. The $S_0 + S_1 + S_2 +$ DOLP combination meets the highest OA of 80.38% and 92.17%, and the $S_0 +$ DOLP combination achieves the second highest OA of 79.51% and 91.82%, respectively. Due to the smaller amount of data, the $S_0 +$ DOLP combination runs about 20 s faster. For dataset A, the $S_0 + S_2$ combination has almost no improvement in accuracy. This is in line with Fig. 4(c) and (i), since only the S_2 images do not clearly show complementarity with S_0. Except for this situation, among all the two-view clustering results, the $S_0 +$ AOP combination performs the worst. The significant noise accompanied by AOP images, as can be seen in Fig. 4, is deemed to be the primary reason for the decrease in accuracy. Similar phenomena can also be observed in the clustering results of the $S_0 + S_1 + S_2 +$ DOLP + AOP combination, which yields 78.22% and 89.37% of OA on two datasets. The addition of AOP resulted in a 2.16% and 2.8% drop in OA on the two datasets, respectively. Overall, the suggested multiview combinations for urban targets of remote sensing scenes are $S_0 + S_1 + S_2 +$ DOLP and $S_0 +$ DOLP. The $S_0 + S_1 + S_2 +$ DOLP combination can achieve the highest accuracy, and the $S_0 +$ DOLP combination has advantages in computing memory and time consumption.

D. Analysis of Tensor Decomposition Methods

Tables II and III have listed the results of various tensorial multiview clustering methods, but the superiority of ITNN in the PHSI clustering task cannot be verified by this. To better compare the performance of different tensor decomposition methods, here, we replace the regularization term $||\mathcal{Z}||_{\text{ITNN}}$ in (12) with other tensorial norms in the proposed TPS-MSC framework. The TNN [29], the t-SVD-based TNN (t-TNN) [31], the weighted TNN (WTNN) [33], and the MLSTD [52] are selected as competitors. The TNN of tensor $\mathcal{Z}$ is defined as the average of the nuclear norm of all the frontal slices of $\hat{\mathcal{Z}}$

$$||\mathcal{Z}||_{\text{TNN}} = \frac{1}{n_3}\sum_{i=1}^{n_3}||\hat{\mathcal{Z}}(:,:,i)||_* \tag{34}$$

where $\hat{\mathcal{Z}}$ denotes the Fourier transform along mode 3 of $\mathcal{Z}$. The t-TNN of tensor $\mathcal{Z}$ is defined as the sum of the singular values of all frontal slices of $\hat{\mathcal{Z}}$

$$||\mathcal{Z}||_{t-\text{TNN}} = \sum_{i=1}^{\min(n_1,n_2)}\sum_{k=1}^{n_3}|\hat{\mathcal{S}}^{(k)}(i,i)| \tag{35}$$

where $\hat{\mathcal{S}}$ comes from the t-SVD of $\hat{\mathcal{Z}}$. The WTNN adds an adaptive weighted vector ω_j, and it is defined as

$$||\mathcal{Z}||_{\text{WTNN}} = \sum_{i=1}^{n_3}\sum_{j=1}^{\min(n_1,n_2)}\omega_j \times \sigma_j\left(\bar{Z}^{(i)}\right) \tag{36}$$

where $\sigma_j(\bar{Z}^{(i)})$ denotes the jth largest singular value of $\bar{Z}^{(i)}$.

The MLSTD decomposes the tensor $\mathcal{Z}$ with the CANDECOMP/PARAFAC model to various rank-1 tensors and introduces a multilayer progressive regularization to measure the sparsity

$$||\mathcal{Z}||_{\text{MLSTD}} = ||\mathcal{Z} - U_1 \circ U_2 \circ \cdots \circ U_N||_F^2 + \sum_{n=1}^{N}\alpha_n||W_n||_{\omega,1}$$
$$\text{s.t. } \nabla_n U_n = W_n V_n^T, \quad V_n^T V_n = 1 \tag{37}$$

where U_n denotes the nth factor matrix of $\mathcal{Z}$, and ∇_n is a linear gradient operation with $[\nabla_n]_{i,i} = 1$ and $[\nabla_n]_{i,i+1} = -1$.

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TABLE V
OA AND RUNNING TIME OF THE TPS-MSC FRAMEWORK WITH VARIOUS TENSORIAL NORMS

Dataset A	TNN	t-TNN	WTNN	MLSTD	ITNN
OA (%)	78.99	79.35	80.15	80.31	80.38
Time (s)	101.6	89.16	103.4	85.60	91.03
Dataset B	TNN	t-TNN	WTNN	MLSTD	ITNN
OA (%)	88.99	91.33	91.11	91.15	92.17
Time (s)	92.53	79.57	95.22	79.30	77.68

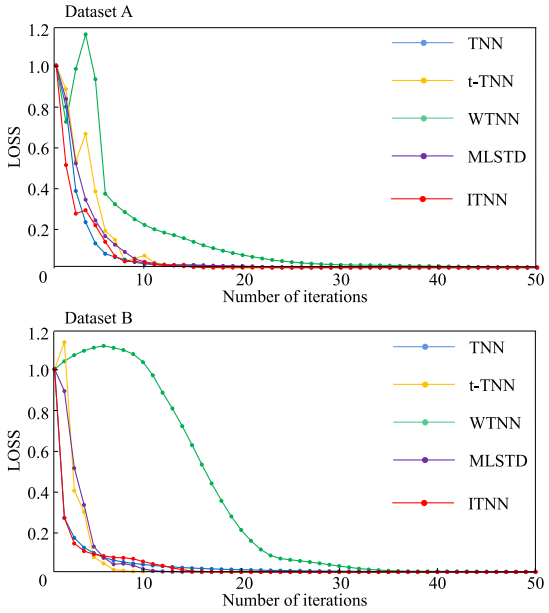


Fig. 7. Convergence curves of the TPS-MSC framework with various tensorial norms.

Table V presents the corresponding clustering results of the TPS-MSC framework with various tensorial norms. To avoid the influence of extraneous factors, all results are implemented with the same dictionary. It can be seen from Table V that all the tensorial norms can achieve a superior performance. Among them, apart from ITNN, which meets the highest OA on both datasets, MLSTD best performed on dataset A and t-TNN best performed on dataset B. All of them, with the TPS-MSC framework, can run much faster than the traditional multiview methods listed in Tables III and IV.

The difference in time consumption can also be observed in Table V. The cause of this difference is associated with the number of iterations required for convergence. Fig. 7 represents the convergence curves of the four tensorial norms. WTNN with the slowest convergence takes the longest running time. By contrast, ITNN adopted in the proposed model occupies the least number of iterations. The experimental results imply that ITNN achieves the best multiview clustering performance most efficiently.

E. Analysis of Parameters

The parameters involved in TPS-MSC will drastically affect the final clustering performance. Since data measured by different instruments in different scenarios may vary in intensity and distribution, it is difficult to find a certain set of parameters

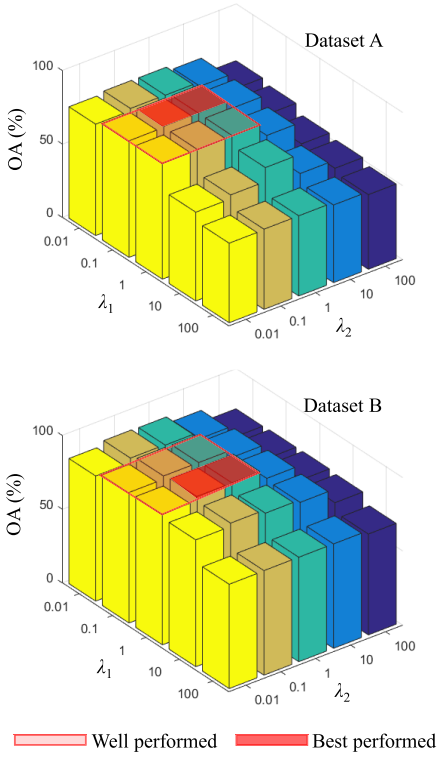


Fig. 8. Analysis of the effect of parameters λ_1 and λ_2.

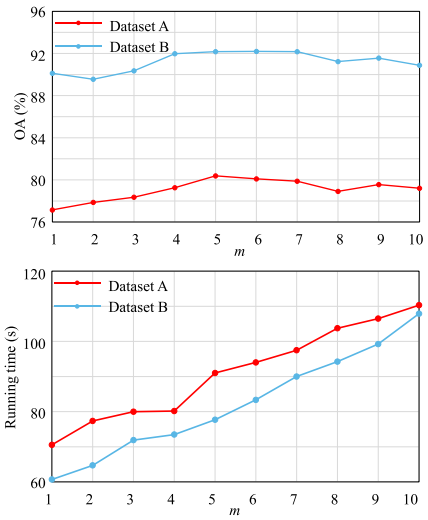


Fig. 9. Analysis of the effect of dictionary size.

that will achieve the best results on different datasets. Thus, the parameters are desired to be hyposensitive to make it easy for users to set them when applying TPS-MSC on a new dataset.

Fig. 8 shows the OAs on two datasets, varying the values of two parameters (λ_1 and λ_2) in the range [0.01, 0.1, 1, 10, 100]. As illustrated in Fig. 8, both parameters show a low sensitivity on two datasets. Dataset A achieves an OA of over 79%, which is deemed to be an acceptable value, within a wild range. Similarly, dataset B achieves over 88% of OA in the same range. The best performed points are deliberately marked on the figures. We can see that setting λ_1 as 0.1 and λ_2

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as [0.1, 1] meets the highest OA on dataset A. On dataset B, the highest OA occurs with λ_1 as 1 and λ_2 as [0.1, 1].

Another parameter that needs to be considered is M, the number of atoms in the dictionary, and it is decided by m, which denotes the number of samples in each cluster for each view. The size of the dictionary directly affects the quality of sparse representation and computational complexity. Fig. 9 shows the OAs and the time consumptions with various values of m. The running time increases monotonically as m increases, and the OA peaks when m takes 5–7.

In summary, the parameters in the proposed method are hyposensitive. The suggested values for λ_1 and λ_2 ranging from 0.5 to 5, and the suggested value for m is 5.

V. Conclusion

In this article, a TPS-MSC algorithm is proposed. To reduce the computational complexity, the TPS-MSC adopts a small size dictionary, which is constructed by independently pre-clustering each view and selecting the data closest to the clustering centers. Under a joint multiview subspace clustering framework, a low-rank tensorial representation exploiting the global complementarity of multiple views is generated from the representation matrices that capture the intraview pairwise relationship. In addition, the multiview pre-clustering consistency constraint is leveraged into the proposed model. Extensive experimental results clearly show that the proposed TPS-MSC algorithm can achieve a superior clustering accuracy with short time consumption and low sensitivity of parameters.

However, the proposed method still has room for further improvement. For example, the cluster number could be automatically determined by subspace detection, and the clustering performance might be further improved with subspace representation by deep learning, which will be in our future work.

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